5 KONVERGENSI CLV DAN KEKONTINUAN BIAYA

Parameter: $a=0, b=2, k=3$

1. Limit Aljabar.

$L_1 = \lim (x \to k) \left[\frac{(x^2-k^2)}{(x-k)}\right], k=3$

$= \lim (x \to 3) \left[\frac{(x-3)(x+3)}{(x-3)}\right]$

$= \lim (x \to 3) [x+3]$ (faktorisasi, $(x-3)$ dikancelkan)

$= 3+3 = 6$

$L_2 = \lim (x \to 0) \left[\frac{\sqrt{(x+(k+1)^2-(k-1)}}{x}\right] \to k+1 = 4$

$= \lim (x \to 0) \left[\frac{\sqrt{(x+16)-4}}{x}\right]$

kalikan konjugat: $\times \frac{[\sqrt{(x+16)+4}]}{[\sqrt{(x+16)+4}]}$

$= \lim_{(x \to 0)} \frac{[(x+16)-16]}{[x(\sqrt{(x+16)+4})]}$

$= \lim_{(x \to 0)} \frac{x}{[x(\sqrt{(x+16)+4})]}$

$= \lim_{x \to 0} \frac{1}{(\sqrt{(x+16)+4})}$

$= \frac{1}{(\sqrt{16}+4)} = \frac{1}{8} = 0{,}125$

$L_3 = \lim (x \to \infty) \frac{[(a+2)x^3+b-x+7]}{(x^3-x+k)}$

$a=0, b=2, k=3$

$\lim (x \to \infty) \frac{[2x^3+2x+7]}{(x^3-x+3)}$

bagi pembilang dan penyebut dgn x^3:

$\lim_{x \to \infty} \frac{\left[\frac{2+2}{x^2} + \frac{7}{x^3}\right]}{\left[\frac{1-1}{x^2} + \frac{3}{x^3}\right]} = \frac{(2+0+0)}{(1-0+0)} = 2$

$L_3 = 2$ (koefisien pangkat tertinggi: $\frac{2}{1} = 2$)

2. Limit Trigonometri

$L_4 = \lim_{(x \to 0)} \frac{\sin((a+1)x) \cdot \tan((b+2)x)}{x^2}$

$a=0, b=2$:

$\lim_{(x \to 0)} \frac{\sin(x) \cdot \tan(4x)}{x^2}$

gunakan: $\lim_{(x \to 0)} \frac{\sin(mx)}{x} = m$ dan

$\lim_{(x \to 0)} \frac{\tan(nx)}{x} = n$

$= \lim_{(x \to 0)} \left[\frac{\sin(x)}{x}\right] \cdot \left[\frac{\tan(4x)}{x}\right]$

$= \lim_{(x \to 0)} \left[\frac{\sin(x)}{x}\right] \cdot 4 \left[\frac{\tan(4x)}{(4x)}\right]$

$= 1 \times 4 \times 1 = 4$

$L_4 = 4$ (teorema: $\lim \frac{\sin(u)}{u} = 1$ dan $\lim \frac{\tan(u)}{u} = 1$ saat $u \to 0$)

3. Limit Tak hingga & konvergensi CLV

d) $R(n) = \frac{(100n+k)}{(n+(a+b+1))}, k=3, a+b+1=3 =$

$= \lim_{(n \to \infty)} \frac{(100n+3)}{(n+3)}$

bagi dgn $n = \lim_{n \to \infty} \frac{(100+3)}{n} / \frac{(1+3)}{n}$

$= \frac{100}{1} = 100$

perdapatan steady-state = Rp 100 jt/bln. Artinya jangka panjang, CLV stabil di angka ini terlepas dr fluktuasi awal.

e) $\lim_{(n \to \infty)} [\sqrt{(n^2+k \cdot n)} - \sqrt{(n^2+b \cdot n)}], k=3, b=2 =$

$= \lim_{(n \to \infty)} [\sqrt{(n^2+3n)} - \sqrt{(n^2+2n)}]$

kalikan konjugat: $\times \frac{[\sqrt{(n^2+3n)}+\sqrt{(n^2+2n)}]}{[\sqrt{(n^2+3n)}+\sqrt{(n^2+2n)}]}$

$= \lim_{(n \to \infty)} \frac{[(n^2+3n)-(n^2+2n)]}{[\sqrt{(n^2+3n)}+\sqrt{(n^2+2n)}]}$

$= \lim_{n \to \infty} \frac{n}{[\sqrt{(n^2+3n)}+\sqrt{(n^2+2n)}]}$

bagi dgn n: $\lim_{n \to \infty} \frac{1}{\left[\sqrt{\left(\frac{1+3}{n}\right)} + \sqrt{\left(\frac{1+2}{n}\right)}\right]}$

$= \frac{1}{(1+1)} = \frac{1}{2}$

Hasil $= \frac{1}{2}$. Interpretasi: selisih proyeksi 2 model CLV yg hampir identik konvergen ke 0,5 jt/bln (perbedaan kecil tp stabil jangka panjang).

4. Kekontinuan Fungsi Biaya Piecewise $C(x)$

Fungsi $C(x)$ dgn 3 potongan di sekitar $x = k = 3$:

$C(x) = \frac{(x^2-p^2)}{(x-p)}, x < p$

$p, x = p$ (atas $x = k = 3$)

$q \cdot x + p, x < p$

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syarat kekontinuan di $x = k$ (3 syarat):

syarat 1 : $C(k)$ ada $\rightarrow C(k) = p$ (sdh didefinisikan)

syarat 2 : $\lim_{(x \to k^-)} C(x) = \lim_{(x \to k^-)} \frac{(x^2 - p^2)}{(x - p)}$

$= \lim_{(x \to k^-)} (x + p) = k + p$

syarat 3 : $\lim_{(x \to k^+)} C(x) = q.k + p$

Agar kontinu : ketiga nilai hrs sama

lim kiri = lim kanan = $C(k)$

$k + p = p \rightarrow k = 0 \leftarrow$ KONTRADIKSI ($k = 3 \neq 0$)

karena $k = p$ (titik sambungan adalah p), maka lim kiri $= 2k$.

Agar kontinu: syarat : $2k = q.k + k \rightarrow k = qk \rightarrow q = 1$

Dan $C(k) = k$ (nilai di titik sambungan)

Jadi : $p = k = 3$, $q = 1 \rightarrow C(x)$ kontinu di $x = 3$

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